

PCA

Problem: FACE RECOGNITION

→ Facial feature analysis using a 3D geometric signature instead of raw pixels.

Feature 1: (x_1) → Eye gap (distance b/w inner corners of eyes in mm).

Feature 2: (x_2) → Nose length (vertical length of nose bridge in mm).

Feature 3: (x_3) → Mouth width (horizontal width of lips in mm).

Sample size = $N = 3$ face images

Target Goal: reduce face representation from 3D to 1D using PCA.

Raw Dataset & Mean Centering

Raw input data vectors:

$$x_1 = \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix} \begin{array}{l} \rightarrow \text{eye gap of face 1} \\ \rightarrow \text{nose length of face 1} \\ \rightarrow \text{mouth width of face 1} \end{array}$$

Similarly for face 2 & face 3:

$$x_2 = \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} \quad x_3 = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$$

Sample Mean Vector (μ): $\mu = \frac{1}{N} \sum_{i=1}^N x_i$

$$\mu = \frac{1}{3} \sum_{i=1}^3 x_i$$

$$\mu = \frac{1}{3} \left(\begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \right) = \frac{1}{3} \begin{bmatrix} 9 \\ 9 \\ 12 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}$$

Average Face of this dataset.

Mean-Centered Deviations: ($x'_i = x_i - \mu$)

$$\begin{cases} x'_1 = \begin{bmatrix} 2-3 \\ 2-3 \\ 5-4 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \\ x'_2 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad \& \quad x'_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \end{cases}$$

→ how each individual face deviates from the baseline norm (shifting the dataset coordinate system center directly to the origin.

Centered Data Matrix (X) :

$$X = [x'_1 \ x'_2 \ x'_3] = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

Covariance Matrix (S)

$$\text{Covariance Formula} = S = \frac{1}{N-1} X X^T$$

$$S = \frac{1}{3-1} X X^T = \frac{1}{2} X X^T$$

$$X X^T = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & -1 & 1 \\ 1 & 2 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$X X^T = \begin{bmatrix} 2 & 3 & -2 \\ 3 & 6 & -3 \\ -2 & -3 & 2 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 1.5 & -1 \\ 1.5 & 3 & -1.5 \\ -1 & -1.5 & 1 \end{bmatrix}$$

main diagonal → variances of
(1, 3, 1)
eye gap, nose length,
mouth width respectively
off diagonal
elements → covariances
among 3 features.

Eigenvalue Calculation

Eigenvalues & Component Ranking

$$\det(S - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 1.5 & -1 \\ 1.5 & 3-\lambda & -1.5 \\ -1 & -1.5 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda_1 = 4.68, \lambda_2 = 0.32, \lambda_3 = 0$$

Variance Percentage Analysis

$$\lambda_1 + \lambda_2 + \lambda_3 = 4.68 + 0.32 + 0 = 5$$

$$\text{Variance captured by } \lambda_1 = \frac{4.68}{5} \times 100 = 93.6\%$$

$$\text{For } \lambda_2 \rightarrow 6.4\%$$

$$\text{for } \lambda_3 \rightarrow 0\%$$

We can drop λ_2 & λ_3 with minimal data loss.

λ_1 contains nearly all of the mathematical information about variations in our database faces.

Eigen Vector (The Primary Eigenface)

$$(S - 4.68I)\vec{v}_1 = \vec{0}$$

$$\begin{bmatrix} -3.68 & 1.5 & -1 \\ 1.5 & -1.68 & -1.5 \\ -1 & -1.5 & -3.68 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

IRREF

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1.78 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{aligned} v_1 + v_3 &= 0 \Rightarrow v_3 = t \\ v_2 + 1.78v_3 &= 0 \Rightarrow v_1 = -t \\ &\Rightarrow v_2 = -1.78t \end{aligned}$$

$$\Rightarrow \vec{v}_1 = \begin{bmatrix} -1 \\ -1.78 \\ 1 \end{bmatrix} \rightarrow \text{Unnormalized vector}$$

Vector normalization (Unit vector Scaling)

$$\|\vec{v}_1\| = \sqrt{(-1)^2 + (-1.78)^2 + (1)^2} = 2.27$$

$$\vec{v}_{\text{normalized}} = \frac{1}{2.27} \begin{bmatrix} -1 \\ -1.78 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.44 \\ -0.78 \\ 0.44 \end{bmatrix}$$

eyegap & nose length scale downward
mouth width increases.
These measure direction only.
(Size/Scale of vector is removed.)

Dimensionality Reduction

Projection Matrix

$$P = [\vec{v}_{\text{normalized}}^T] = \begin{bmatrix} -0.44 & -0.78 & 0.44 \end{bmatrix}$$

→ 1D

Projection of Face 1 into 1D eigenspace

$$y_1 = P x_1'$$

$$y_1 = \begin{bmatrix} -0.44 & -0.78 & 0.44 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

mean centered vector of face 1

$$y_1 = 1.66$$

→ 3D vector of face 1 is compressed into a single scalar score of 1.66

face 2

$$y_2 = \begin{bmatrix} -0.44 & -0.78 & 0.44 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = -2.44$$

face 3

$$y_3 = 0.78$$

feature deviate from average face in the opposite direction of face 1.

Interpretation

If $y_i = 0 \rightarrow$ face is average. (it matches the average face we calculated)

If $y_i > 0 \rightarrow$ face shifts with the direction of eigenface vector.

If $y_i < 0 \rightarrow$ face shifts in the exact opposite direction of eigenface vector.

Primary Eigenface Direction :

Rule Vector = $\begin{bmatrix} -0.44 \\ -0.78 \\ 0.44 \end{bmatrix}$ $\begin{matrix} \rightarrow \text{Eye Gap Rule} \\ \rightarrow \text{Nose Length Rule} \\ \rightarrow \text{Mouth width Rule} \end{matrix}$

Face1

$y_1 = 1.66 \rightarrow$ follows the rule vector directly

$\begin{matrix} \rightarrow \text{wider mouth than average} \\ \rightarrow \text{narrower eye gap than avg.} \\ \rightarrow \text{shorter nose than avg.} \end{matrix}$

Face2

$y_2 = -2.44 \rightarrow$ flips the rule vector entirely

$\begin{matrix} \rightarrow \text{narrower mouth than avg.} \\ \rightarrow \text{wider eye gap} \\ \rightarrow \text{longer nose} \end{matrix}$

Face3

$y_3 = 0.78 \rightarrow$ follows the rule vector slightly.

0.78 is closer to 0
Face 3 is much closer to avg. face.